

AI PROBLEM SCOPING

Hostel Room Prediction System

A structured 8-step scoping framework designed for translating an institutional accommodation request into a measurable, interactive, and predictive machine-learning solution.



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STEP AGENDA

Eight steps from administrative request to AI deployment solution:

- **01** Understanding the Business Problem
- **02** Defining the Problem Precisely
- **03** Is AI Really Needed?
- **04** Identifying Stakeholders
- **05** Defining the AI Task
- **06** Defining the Success Metrics
- **07** Defining Constraints
- **08** Assumptions and Risks

01. UNDERSTANDING THE BUSINESS PROBLEM

Initial Statement: Hostel room prediction system

Core Problem Context: Hostel administration cell wants to predict how many students will take hostel rooms this year on behalf of the previous year's data. The administration cell wants to ensure there is no shortage of rooms this year. They approach an AI team/organization and ask to build an AI model to predict the hostel rooms on behalf of how many students take hostel rooms.

Stakeholder Inquiries formulated by the AI Team:

Academic Dept.

- Previous year admissions?
- Current year admissions?

Administration

- Hostel room trends?
- AC vs Non-AC preferences?

Hostel Warden

- Curfew rules & entry timings?



STAKEHOLDER FEEDBACK & INSTITUTIONAL ANSWERS

The administration cell reverts the following factual findings:

- **Previous Year Admissions:** 3000 students took admission
- **Current Year Admissions:** 4000 students took admission
- **Previous Year Hostel Allocation:** 300 students took hostel rooms
- **Current Year Hostel Allocation:** 400 students took hostel rooms
- **Room Preference Pattern:** Double sharing room is more preferred by students
- **Fee Structure Modifications:** Yes, the price of all rooms is less by 1000 in the current year
- **Hostel Timings Rules:**
 - Hostel Entry Timing: 7:00 PM | Hostel Exit Timing: 8:00 AM
 - Curfew Timing Restriction: Students are not allowed in the hostel after 7:30 PM

Previous Year Fee Structure Matrix

Room Type	Without AC Fee	With AC Fee
Single Bed	3000 / month	4000 / month
Double Bed	5000 / month	6000 / month
Triple Bed	7000 / month	8000 / month

Current Year Fee Structure Matrix (with 1000 price drop)

Room Type	Without AC Fee	With AC Fee
Single Bed	2000 / month	3000 / month
Double Bed	4000 / month	5000 / month
Triple Bed	6000 / month	7000 / month

02. DEFINING THE PROBLEM PRECISELY

POOR PROBLEM STATEMENT

"Hostel room prediction system"

BETTER APPROACH

"Identify how many students take hostel in next year on the basis of admission and rules and fee structure so that administration cell can give more facility to it and prevent room shortage"

The Precision Mapping (5 Ws Framework)

Who?	What?	When?	Why?	Outcome?
Students	Predict hostel rooms	Before the admission session	So that administration cell can give more facility to it	Administration cell can prevent shortage

03. IS AI REALLY NEEDED?

Core Question: Can the problem be solved without AI?

Manual Rule Constraint: If a student lives far from campus around 35 Km, then the student always takes a hostel room.

Why use Machine Learning (ML)?

AI should only be used when patterns are complex, multiple factors interact, and manual rules are insufficient. In this use case, a student taking a hostel is dependent on the following factors:

- Fee structure
- Financial status
- Rules and restrictions
- Cleanliness
- Facilities provided
- General environment

Conclusion: Pattern is going to be complex → AI is needed

04. IDENTIFYING STAKEHOLDERS

Stakeholder Mapping Matrix:

- **Administration Cell:** Prevent shortage of rooms
- **Student:** Take facility / secure accommodation
- **Parents:** Student success and safety
- **Campus:** Gain dynamic institutional profit
- **Faculty:** Ensure students reach the class on time

05. DEFINING THE AI TASKS

Machine Learning Formulation:

- **Classification Problem (Binary Classification Target):**
Output → *Take Hostel Room = Yes / Take Hostel Room = No*
- **Risk Scoring & Likelihood Estimation:**
How likely will a student take a hostel room? Range: *0 - 100%*
- **Ranking Problem / Incentive Optimization:**
Identify the top 200 students who will take a hostel room to give a 10% discount.

06. DEFINING THE SUCCESS METRICS

- **Technical Metrics:** Accuracy, Precision, Recall, F1 Score, ROC-AUC.

07. DEFINING CONSTRAINTS

- **Operational Constraints:** Administration cell can only give 10% discount to the top 200 students.
- **Time Constraints:** Models must finalize execution before starting the next admission session.
- **Data Constraints:** Mental health indicators, family circumstances, updated financial records.
- **Ethical Constraints:** Students should not be harmed, personal data safety, no discrimination.

08. ASSUMPTIONS AND RISKS

Assumptions

- Financial records are strong.
- Discipline is mandatory.
- Discounts strictly on proper recommendation.

Risks

- Missing data fields
- Biased training decisions
- Sudden changes in academic policies

Thank you.

Questions on hostel room prediction scoping, algorithm targets, or capacity optimization frameworks — happy to discuss.